

# Arshia Soori

Backend & AI Tech Lead | Senior Python Engineer


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**Hands-on backend and AI tech lead** with 5 years of technical leadership and 4+ years owning product architecture end to end. On the backend I design Django/DRF and Django Ninja microservices and event-driven systems (Celery, RabbitMQ, Redis, MQTT) that handle sensitive data reliably at scale. On the AI side I build LLM agents, RAG pipelines, and ML models that turn raw wearable and radio sensor data into meaningful insights. As CTO of two knowledge-based products, I lead and hire engineering teams, set the engineering standards (code review, branching and release, testing, API documentation), and turn business requirements into reliable, observable products.

## Skills

### I already know

Python 5/5 | Django 5/5 | Django REST Framework 5/5 | Django Ninja 5/5

FastAPI 4/5 | LangChain 4/5 | LlamaIndex 3/5 | Pydantic AI 4/5 | Pydantic 3/5

PostgreSQL 4/5 | Celery 4/5 | RabbitMQ 3/5 | Redis 4/5 | MQTT 4/5

Hugging Face 3/5 | PyTorch 3/5 | TensorFlow 3/5 | Docker 4/5 | Kubernetes 2/5

Linux 4/5 | Git 5/5

### I also work with

OpenAPI / Swagger | Keycloak | GitLab CI | GitHub Actions | ELK Stack | Prometheus | Grafana

### I speak

Persian native | English fluent

## Work experience

### CTO and Hands-on Tech Lead

March 2026 - now

DelBand

I hereby give consent for my personal data included in my application to be processed for the purposes of the recruitment process.

Iran's first smart health monitoring service and the first holder of a Med 3.0 license, built by Ayandeh Pardaz, a technology company backed by Hamrahe Aval (MCI) and its investment fund Harekat Aval. Wearable rings and watches stream sensor data (ECG, HRV, sleep, falls, blood oxygen) that is analyzed and shown in the app, with emergency alerts to family members and caregivers.

- Lead a team of 4 engineers and own architecture, technical strategy, hiring, and delivery.
- Started the backend as a Django/DRF monolith, then led its refactor into microservices built on Django Ninja.
- Run alarms, notifications, OTP, and background health analysis as asynchronous jobs on Celery, RabbitMQ, and Redis, with retries, circuit breakers, and idempotent processing so emergency alarms are neither lost nor sent twice.
- Lead a team dedicated to medical algorithms, turning raw wearable sensor data into health insights with ML and AI.
- Building AI agents that answer users' questions about their health data and give exercise, food, and sleep recommendations.
- Designed PostgreSQL and TimescaleDB schemas for high-volume sensor time series, and secured health data with Keycloak, role-based access control, and encryption.
- Defined engineering standards across DelBand and IndoorA: code review rules, branching and release process, testing requirements, and OpenAPI documentation.
- Set up CI/CD, Docker and Kubernetes deployments, and observability with ELK, Prometheus, and Grafana.

**Technologies:** Python, Django, Django Ninja, Celery, RabbitMQ, Redis, PostgreSQL, TimescaleDB, LangChain, LlamaIndex, Pydantic AI

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### CTO and AI Tech Lead

September 2022 - February 2026

IndoorA

The first indoor positioning and navigation system in the Middle East, built by Ayandeh Pardaz and serving asset tracking, workforce safety, and visitor wayfinding for enterprise and public venues.

- Led a team of 6 engineers from early product to a certified knowledge-based platform, and hired 3 engineers across both products.
- As AI tech lead, led development of the positioning engine fusing BLE, Wi-Fi, and UWB signals with KNN, neural networks, CNNs, LSTMs, Kalman filters, and particle filters.
- Designed a microservice architecture with DRF and ADRF (async DRF) services, plus a dedicated Django Ninja edition for Tehran International Exhibition Center.
- Built event-driven location processing over MQTT and RabbitMQ, with Celery workers for background position calculation and safety alerts.
- Delivered and integrated the platform for MCI, Radin, Tehran International Exhibition Center, Iran Mall, and Mobarakeh Steel Company.

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**Technologies:** Python, Django REST Framework, Django Ninja, MQTT, RabbitMQ, Celery, PostgreSQL, PyTorch, TensorFlow, Hugging Face, GPU (CUDA)

### Tech Lead

September 2021 - August 2022

**Advanced Aerospace Structures Laboratory, K. N. Toosi University of Technology**

Led the development of an intelligent entry/exit management system combining image processing, CNNs, and language models.

- Led a team of 7 engineers, setting technical direction and reviewing implementation.
- Built real-time CNN-based (YOLO) detection of vehicles and people, with automated door control, guest management, and an operator reporting dashboard.
- Used language models to generate human-readable activity summaries and reports.

**Technologies:** Python, OpenCV, YOLO, PyTorch, GPU (CUDA)

## Projects

### DelBand

March 2026 - now

Iran's first smart health monitoring service, and the first to hold a Med 3.0 license. Owned and operated by Ayandeh Pardaz, developed in collaboration with and supported by MCI, backed by MCI's investment fund Harekat Aval.
Wearable devices (rings, watches) capture sensor signals which are processed and turned into meaningful health insights shown in the app. AI agents answer users' questions about their data, help them understand it, and give recommendations on exercise, food, and sleep plans. For elderly users, the platform tracks ECG, HRV, sleep, falls, blood sugar, and blood oxygen, and can send notifications and emergency calls to observant family members or caregivers.

**My role:** CTO

**Company:** Ayandeh Pardaz

**Backed by:** Hamrahe Aval (MCI), Harekat Aval

**Category:** Health tech, AI agents

**Recognition:** Certified knowledge-based product, Selected Health & Care project, Presidential Innovation and Prosperity Fund, 6th International mHealth Congress of Shiraz University of Medical Sciences

**Technologies:** Python, Rust, Django, Django Ninja, Pydantic AI, LangChain, LlamaIndex, Hugging Face, PostgreSQL, TimescaleDB, GPU (CUDA)

### IndoorA

September 2022 - February 2026

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The first indoor positioning and navigation system in the Middle East, owned and operated by Ayandeh Pardaz. A single knowledge-based positioning and navigation engine (BLE, Wi-Fi, UWB, beacons, and gateways with KNN, neural networks, CNNs, LSTMs, Kalman filters, and particle filters) powering three business models:

- Indoor positioning for organizations managing physical and human assets: employee productivity, asset inventory, access control, and guest movement management. (Radin, MCI)
- Life safety: continuous monitoring of access levels, movement through hazardous areas, and incidents, sending alerts to rescue teams and managers. (Mobarakeh Steel Company)
- Positioning, wayfinding, visitor management, and personalized offers in malls and exhibitions, connecting business owners with their customers. (Tehran International Exhibition Center, Iran Mall)

**My role:** CTO and AI Tech Lead

**Company:** Ayandeh Pardaz

**Category:** Indoor positioning, Navigation

**Recognition:** Member of Iran's Knowledge-Based Companies

**Technologies:** Python, Rust, Django, Django Ninja, PostgreSQL, TimescaleDB, Bluetooth / BLE, Wi-Fi, TensorFlow, PyTorch, Hugging Face, MQTT, GPU (CUDA)

### Intelligent Traffic and Movement Management System

September 2021 - August 2022

An intelligent traffic and movement management platform combining computer vision and language models to automate entry/exit control. CNN-based image processing detects and identifies vehicles and people in real time, driving automated door open/close control and streamlined guest management. Language models interpret activity and generate human-readable summaries, surfaced through a dashboard with detailed reports for facility operators.

**My role:** Tech Lead

**Company:** Advanced Aerospace Structures Laboratory, K. N. Toosi University of Technology

**Category:** Computer vision, AI

**Technologies:** Python, OpenCV, YOLO, PyTorch, TensorFlow, Jupyter, Kaggle, GPU (CUDA)

## Education

### Aerospace Engineering

September 2020 - July 2024

K. N. Toosi University of Technology

Bachelor's degree in Aerospace Engineering.

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